MATH ~~11/12/22~~ 23/12/2023

1. $a_{100} = ?$ $d = \sqrt{3}$ $a = \sqrt{3}$

$a_1 = \sqrt{3}$ $a_2 = 2\sqrt{3}$, $3\sqrt{3}$

$$a_n = a + (n-1)d$$
$$a_{100} = \sqrt{3} + (100-1)\sqrt{3}$$
$$a_{100} = \sqrt{3} + (99\sqrt{3})$$
$$\boxed{a_{100} = 100\sqrt{3}}$$

2. Infinitely many.

3. $\cancel{S_n =}$ $S_{28} = ?$

$n = 8n - 5$

$$S_{28} = 8(28) - 5$$
$$S_{28} = 140 - 5$$
$$8\boxed{S_{28} = 135}$$

5. $a = 3$

$d = 11$

$a_{54} = a + 53d$

$a_{54} = 3 + (53 \times 11)$

$= 3 + 583$

$\boxed{a_{54} = 586}$

$132 + 586$

$= 718$

$718 = a + (n-1)d$

$718 = 3 + (n-1)11$

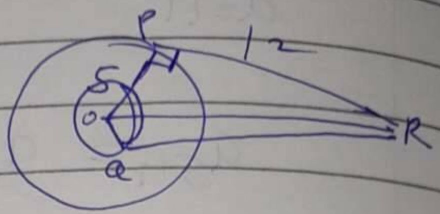
$718 - 3 = (n-1)11$

$\frac{715}{11} = n-1$

$65 + 1 = n$

$\boxed{66 = n}$

6. PR = 12 OP = 5 OQ = 4



In ΔROP

~~OP⊥~~ $\angle OPR$ is 90° (radii $\perp$ on PR)

by Pythag.. :

$\cancel{(PR)^2 + \ldots = OP}$

$(OR)^2 = (PO)^2 + (PR)^2$

$= (5)^2 + (12)^2$

$= 25 + 144$

$(OR)^2 = 169$

$\boxed{OR = 13}$

$\cancel{12^2}$ $\cancel{144}$

$\cancel{13^2}$ $\cancel{169}$

7. 174, 261, ... 957

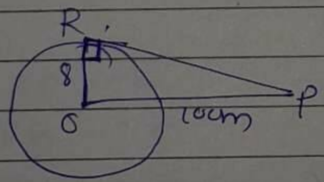
$a = 174$. $d = 87$.

$a_n = 957$

$957 = a + (n-1)d$
$957 = 174 + (n-1)87$
$957 - 174 = (n-1)87$
$\frac{783}{87} = n-1$

$9 = n-1$
$9+1 = n$
$\boxed{10 = n}$

8



RO ⊥ to PR (radii ⊥ on tangent)
In △PRO
by ~~pytho~~ pytha
$\cancel{(OP)^2 = O}$ $(OP)^2 = (OR)^2 + (RP)^2$
$(10)^2 = (8)^2 + (RP)^2$
$100 = 64 + (RP)^2$

$100 - 64 = (RP)^2$

$36 = \cancel{RP} (PR)^2$

$\boxed{6 = PR}$

9. $S_{51} = ?$

$a_2 = 14 \quad a_3 = 18$

$d = 4$

$\cancel{S_{51} = \frac{n}{2}(2a + (n-1)d)}$

$= \frac{51}{2}(2c$

$a_2 = a + d$

$14 = a + 4$

$14 - 4 = a$

$\boxed{10 = a}$

$S_{51} = \frac{n}{2}(2a + (n-1)d$

$= \frac{51}{2}(2(10) + (51-1)4$

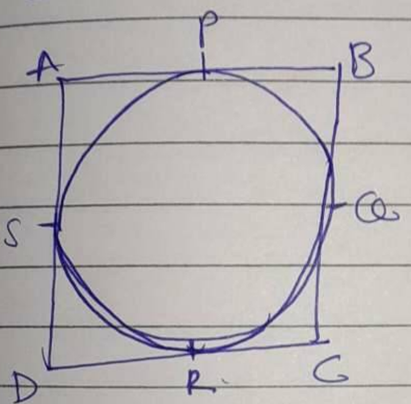
$= \frac{51}{2}(20 + (50)4)$

$= \frac{51}{2}(20+200)$

$= \frac{51}{\cancel{2}} \times \overset{110}{\cancel{220}}$

$\boxed{S_{51} = 5610}$

10.



Given: AB, CB, RC & AD are tangent to the circle at Pt P, Q, R & S

$\square ABCD$ is parallelogram

AB = CD & AD = BC

AP = SA
PB = QB
CR = CQ
RD = DS

AP + PB + CR + RD = SA + QB + CQ + DS
AB + DC = BC + AD.
AB + AB = BC + BC
$\cancel{2}AB = \cancel{2}BC$

$AB = BC$

$\therefore AB = BC = AD = AC$.

$\therefore$ it's a rhombus.